

# Laboratory development of a LIDAR for measurement of sound velocity in the ocean using Brillouin scattering

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## ABSTRACT

There have been several studies of the potential accuracy of LIDAR measurements of sound velocity in the ocean by measuring the spectral shift of the backscattered Brillouin lines. However, due to technical limitations, such systems have not previously been experimentally demonstrated. Measurement of the Brillouin shift as a function of depth in the ocean requires a stabilized, narrow linewidth, pulsed laser, and a high-resolution spectroscopic technique. We have used a scanning Fabry-Perot to obtain the first frequency resolved measurements of Brillouin scattering in water using a pulsed laser; these results will be presented here. But for the practical application of measuring Brillouin shifts as a function of depth in the ocean a non-scanning spectroscopic technique is required to measure the small frequency shifts; the edge technique meets this requirement. Using it in conjunction with the edges of absorption lines in the molecular spectra of  $I_2$  and/or  $Br_2$ , avoids the limitations associated with use of a Fabry-Perot etalon; specifically, its small solid angle of acceptance and its vulnerability in noisy environments. This new approach will be briefly described.

**Keywords:** Brillouin LIDAR remote sensing, sound velocity, ocean, edge technique, molecular absorption lines.

## 1. INTRODUCTION

Several studies have discussed the importance of remote measurements of sound velocity in the upper-ocean mixed layer.<sup>1-5</sup> Real-time range-resolved monitoring provides a crucial input to the understanding of the physical behavior of the ocean. A Brillouin LIDAR is a promising tool for such remote measurements of the sound velocity as a function of depth. Brillouin scattering produces two inelastically scattered Lorentzians centered symmetrically with respect to the laser line; they have a FWHM of about 0.5 GHz. The so-called Brillouin shift  $\nu_B$ , that is to say the frequency shift between the laser line and each of the Brillouin lines, is typically between 7 and 8 GHz, and is proportional to the sound velocity. In pure water, the scattering spectrum is almost solely composed this doublet. An elastically scattered central line (also called the unshifted line, or improperly the Rayleigh line)<sup>6</sup> appears when the water is contaminated by hydrosols.

The limits in the accuracy of sound velocity and temperature profiles in the ocean from LIDAR Brillouin shift measurements have been discussed in detail by Fry, *et al.*<sup>7</sup> Briefly, for a given incident laser wavelength  $\lambda$  the Brillouin shift  $\nu_B$  is proportional to the refractive index of water  $n$  and the sound velocity  $v_s$ . The refractive index itself is function of the salinity  $S$  and temperature  $T$ . In addition to these two variables, the sound velocity also has a weak dependence on pressure, but for present purposes, this dependence will be neglected; the generalization to include pressure dependence is straightforward. The Brillouin shift is then expressed as:

$$\nu_B(S, T) = \frac{2n(S, T)}{\lambda} v_s(S, T) \sin(\theta/2), \quad (1)$$

where  $\theta$  is the scattering angle ( $\sin(\theta/2)=1$  for  $180^\circ$  backscattering). Clearly, there are five parameters ( $\nu_B$ ,  $v_s$ ,  $n$ ,  $S$ , and  $T$ ), but there are only three relations between them and only one measurement ( $\nu_B$ ). The three relations are Eq. 1 and empirical equations for  $n(S, T)$  and  $v_s(S, T)$ . Thus one additional piece of information is required. For present purposes, it is provided by the means and standard deviations of the extensive set of salinity measurements (from 1900 to 1990) compiled by the National Oceanic and Atmospheric Administration.<sup>8</sup> With these salinity estimates based on historical data, i.e. without actual measurements of salinity, the limits on the sound velocity accuracy can be better than 0.3 m/s.

Our primary interest is in developing a LIDAR for range-resolved measurements of sound velocity in the ocean via the measurement of the Brillouin shift. Although the idea of a Brillouin LIDAR is not new,<sup>1-3</sup> such systems have not yet been experimentally demonstrated due to technical limitations. Measurements of the Brillouin shift as a function of depth in the ocean requires both a stabilized narrow linewidth pulsed laser, and a high-resolution non-scanning spectroscopic technique.

Sound velocity measurements with depth resolution  $\delta z$  of 1 meter require a Brillouin frequency shift measurement for each successive 10 ns ( $=2n\delta z/c$ ) interval of the LIDAR return. In addition, a precision of 1 m/s for the sound velocity requires the measurement of Brillouin shifts with an accuracy better than about 4 MHz.<sup>7</sup> This small value compared to the linewidth of the Brillouin line (500 MHz) does not allow the use of a conventional spectrometer. In this paper we discuss our recent progress in development of a Brillouin LIDAR.

The first laboratory measurements of frequency resolved Brillouin lines using a pulsed laser and a scanning Fabry-Perot technique for frequency discrimination are presented in Sect. 2. Previous pulsed laser measurements<sup>2</sup> were not completely resolved, but our results clearly demonstrate this resolution with pulsed lasers. However, the scanning Fabry-Perot measurement technique used to obtain this data is not accurate and fast enough to be used directly for LIDAR measurements of the Brillouin shift. Nevertheless, these results and the measurement apparatus will provide a valuable reference for the initial data obtained with our new non-scanning spectroscopic technique.

This new technique is a version of the edge technique, and is a powerful non-scanning spectroscopic technique for the measurement of small frequency shifts.<sup>9</sup> It has numerous applications, and has been successfully used in a Doppler LIDAR for atmospheric wind velocity measurements.<sup>10</sup> From the standpoint of an oceanic Brillouin LIDAR, the main limitations of previous implementations of this method are the small acceptance angle of the Fabry-Perot etalons used for the frequency discrimination, and the difficulties associated with operating these interferometers in noisy vibrational and acoustic environments. We are implementing a version of the edge technique in which the frequency discriminants are the edges of absorption lines in the molecular spectra of  $I_2$  and/or  $Br_2$ . Explicitly, the frequency of the laser transmitter is chosen so that the edges of two different molecular absorption lines (with opposite slopes) are located at the centers of the Brillouin lines. Because of the steep slope of the absorption edges, small changes in the Brillouin shift produce large changes in transmission. The details of the experimental set up are described in Sect. 3.

## 2. FREQUENCY RESOLVED MEASUREMENTS OF BRILLOUIN LINES

The theoretical model for Brillouin scattering in a liquid was developed by Mountain<sup>4</sup> and Fabelinskii.<sup>11</sup> Early attempts to examine the use of Brillouin scattering for the determination of sound velocity in pure water were by Chiao, *et al.*<sup>12</sup> and by Benedek, *et al.*<sup>13</sup> Since then, considerable effort with CW lasers has been reported,<sup>5</sup> but no definitive results have previously been performed using a pulsed laser. To our knowledge, the only such published results show a FWHM of the Brillouin lines of about 3 GHz; this indicates that the stability and linewidth of the laser transmitter were not sufficient.<sup>2</sup>

Our experimental set up for the Brillouin spectrum measurements is shown in Fig. 1. A frequency doubled, injection seeded, Q-switched, pulsed Nd:YAG laser provides 75 mJ at 532 nm, with a repetition rate of 10Hz. Using an injection locking technique developed in our laboratory,<sup>14</sup> it operates at a single frequency and has stable Fourier transform limited pulses of 10 ns, i.e. a frequency bandwidth of about 50 MHz. A horn shaped water cell is used to avoid backreflections. An additional light baffle consisting of two pierced parallel plates was added in the bottom of the water cell to further suppress parasitic backreflections. Noise due to the reflection from the surface of the water is avoided by using another thin baffle plate. The backscattered light is collected at an angle of 175° relative to the laser beam direction. The acceptance angle of the optical system is  $\approx 1^\circ$  and is determined by the focal length and the size of lens L1; this acceptance angle does not significantly increase the FWHM of the Brillouin lines. The most critical point in order to obtain good frequency discrimination is to provide a very well collimated beam to the input of the Fabry-Perot (FP) etalon. A spatial filter consisting of 2 lenses (L1 and L2), a pinhole (P1) of 10  $\mu$ m diameter, and one iris (I) is used for this purpose. To avoid background radiation, the output of the etalon is focused through pinhole P2 onto a photomultiplier, PM, and the overall receiver system is enclosed in a black box.

The Fabry-Perot has a resolution of 0.25 GHz ( $R=96\%$ ) and a Free Spectral Range (FSR) of 18 GHz. The etalon piezoelectric transducers are driven by a Burleigh RC-42 ramp generator. The measurements are performed with a scanning ramp duration of 2 minutes per FSR. The detector is a Hamamatsu R446 photomultiplier tube operating in the analog mode. The signal is integrated and averaged over 10 laser shots by a gated integrator and boxcar averager. It is digitized and stored using a Labview program on a MacII computer.

Fig. 2 shows a measured frequency spectrum, superposed on the theoretical one for a water temperature of 29°C (corresponding to a Brillouin shift of 7.55 GHz). Only one free spectral range of the Fabry-Perot is presented in the figure. The experimental points are raw data from a single scan; no further processing has been performed. There is very good

agreement with theory. The strong central peak is due to particulate contamination of the water, which is apparently due mostly to oxidation of the metal light baffle submerged in the horn shaped water cell.

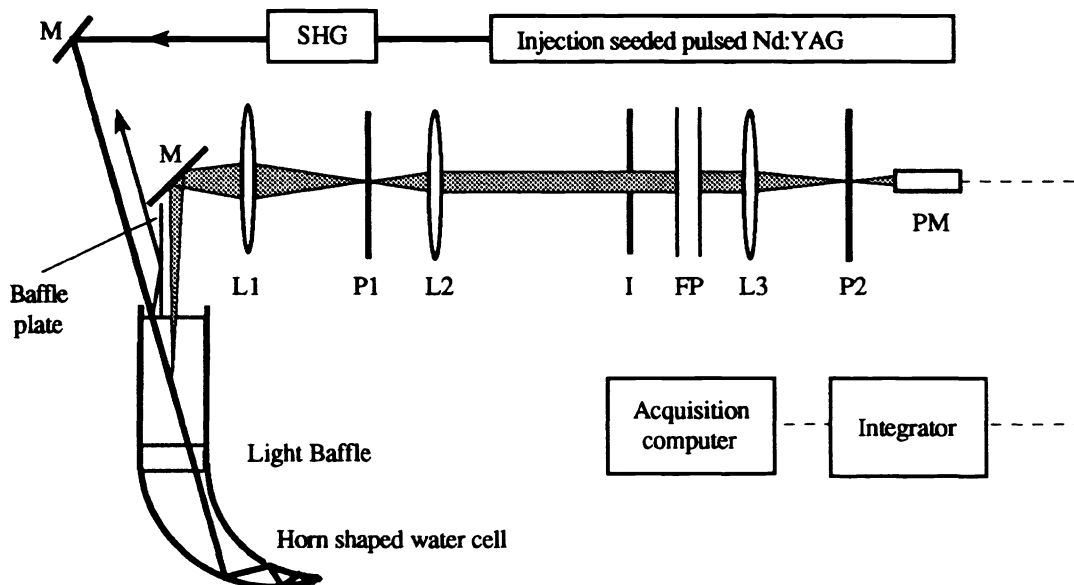


Fig. 1. Brillouin spectrum measurement set up. L1, L2, and L3 are biconvex lenses of focal length 50 cm, 20 cm and 8 cm, P1 is a 10  $\mu\text{m}$  diameter pinhole, I is an iris used to select the lower interference order, P2 is a 1 mm diameter pinhole, SHG is second harmonic generator, M are mirrors, and PM is the photomultiplier.

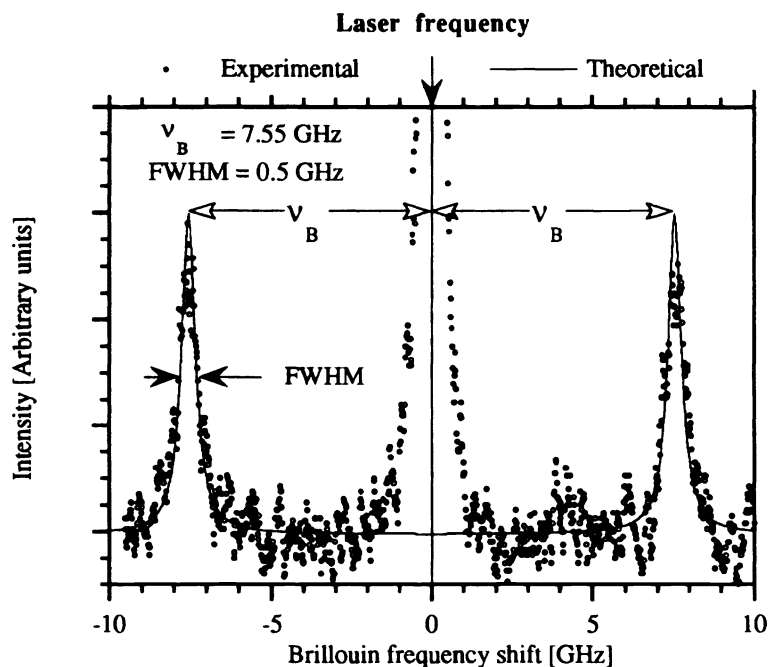


Fig. 2. Experimental (dots) and theoretical Brillouin spectrum of water at 29°C using the second harmonic of a pulsed injection seeded Nd:YAG laser (532 nm).

### 3. BRILLOUIN SHIFT MEASUREMENT

As discussed in the introduction, a Brillouin LIDAR based on a version of the edge technique is under development. Using the edge technique, atmospheric measurements at 1064 nm have shown wind velocity accuracies of 0.25 m/s, corresponding to frequency changes smaller than 470 kHz.<sup>10</sup> This is the most sensitive method yet demonstrated for small frequency shift LIDAR measurements.<sup>15</sup> Versions of the edge technique had already been considered by Hirschberg, *et al.*<sup>3</sup> and by Hickman, *et al.*<sup>1</sup> for oceanic Brillouin shifts measurements using a laser wavelength of 532 nm. In the latter study the statistical uncertainty  $\delta v_B$  in the measurement of the Brillouin shift is calculated assuming a single laser shot with an energy of 500 mJ and a depth resolution of 1.1 meter. For  $\gamma < 0.04 \text{ m}^{-1}$ , shot noise limited uncertainties in  $\delta v_B$  of less than 1 MHz can be expected for depths down to 75 meters, and of less than 4 MHz down to 100 meters. These correspond to sound velocity measurement accuracies that can be better than 0.3 m/s and 1 m/s,<sup>7</sup> respectively.

We are developing a version of the edge technique that makes use of the edges of absorption lines in the molecular spectra of  $\text{I}_2$  and/or  $\text{Br}_2$ , as opposed to the Fabry-Perot interferometers used in other applications<sup>1-3,9</sup> of the edge technique. The schematic of such a LIDAR receiver is shown in Fig. 3. The second harmonic of an injection-seeded pulsed Nd:YAG laser is used as the transmitter. The laser signal backscattered by the ocean water is collected by a telescope, and is first directed through an atomic cell (used as a blocking-filter) where the central line of the backscattered frequency spectrum is absorbed. The backscattered signal transmitted by the blocking filter is then split into two equal parts: one part is detected to produce a normalization signal (S2). The other part passes through the edge filter and is detected to produce the signal (S1). The Nd:YAG laser frequency stabilization is performed by picking off a small part of the cw seed laser beam, frequency doubling it in an external ring build-up cavity, and locking the seed laser to the absorption line of the blocking-filter. The return signals S1 and S2 are sampled every 2-3 ns to produce a range-resolved measurement (of course, the temporal resolution is limited by the 10 nanosecond laser pulse).

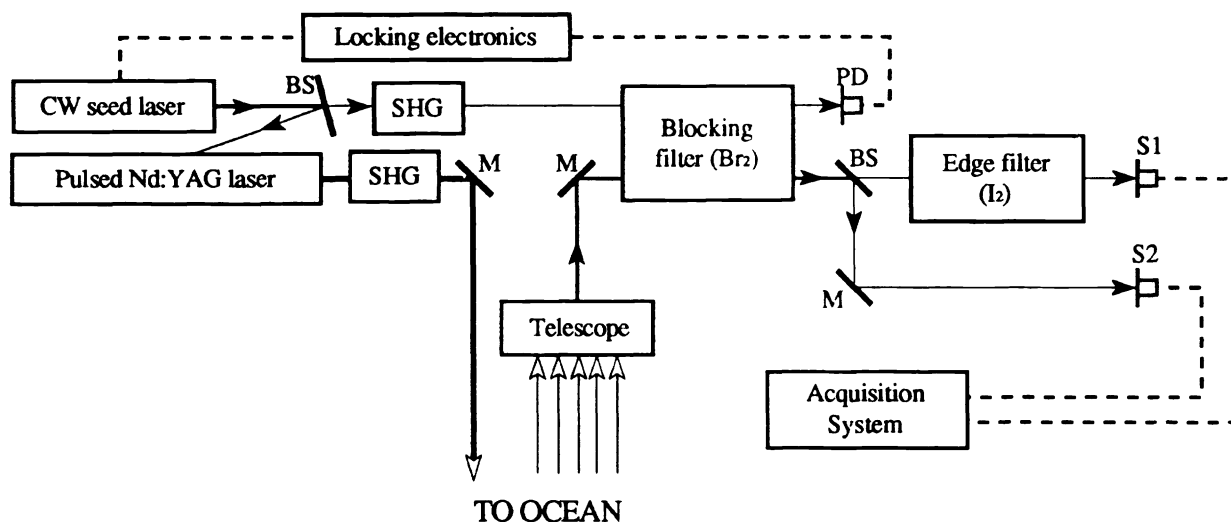


Fig. 3. Brillouin LIDAR receiver schematic. The Brillouin shift is determined from the ratio  $S1/S2$ . BS are beam splitters, M are mirrors, SHG is second harmonic generators, M are mirrors, and PD are photo detectors. Solid lines correspond to light paths, and dotted lines to electrical connections.

As an example of an edge filter, consider a  $^{127}\text{I}_2$  absorption cell. For a laser wavelength of 532.573806 nm, it provides two appropriately positioned absorption edges (see Fig. 4). Because of the steep average slope of the absorption edges, small changes in the Brillouin shift produce large changes in the transmission of the edge filter. The temperature of the gas and/or the vapor pressure in the cells can be changed for a fine adjustment of the absorption line slopes and frequencies.

An important advantage of this edge filter version is that a change in the incident laser frequency or any Doppler shifts due to movements will increase the edge filter signal corresponding to one Brillouin line and decrease the signal corresponding to the other; thus such systematic effects are canceled to first order. The ratio  $S1/S2$  provides a

normalization which makes the frequency discrimination independent of the intensity of the light backscattered by the ocean water, i.e. independent of the attenuation due to water absorption or hydrosols scattering.

An ideal blocking filter should exhibit total absorption for the central line of the backscattered spectrum, and good transmission for the Brillouin peaks. The FWHM of the central line is mainly the linewidth of the laser transmitter, and is  $\approx 50$  MHz. Fig. 5 shows the absorption spectrum of  $\text{Br}_2$  superposed on theoretical Brillouin lines with a shift of 7.5 GHz, and for the same incident laser wavelength as for the  $\text{I}_2$  data in Fig. 4. This absorption line provides a satisfactory, although not perfect blocking filter. Of course, the convolution of the Brillouin lines and of the transmission spectrum of  $\text{Br}_2$  has to be considered as the input to the edge filter.

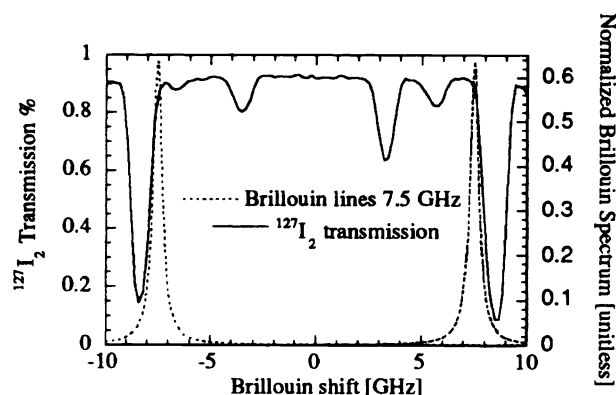


Fig. 4. Edge filter: theoretical Brillouin lines for a frequency shift of 7.5 GHz superposed on the absorption spectrum of  $^{127}\text{I}_2$ . Small changes of the Brillouin shift make large changes in the transmitted signal. The incident wavelength for the Brillouin spectrum is 532.573806 nm.

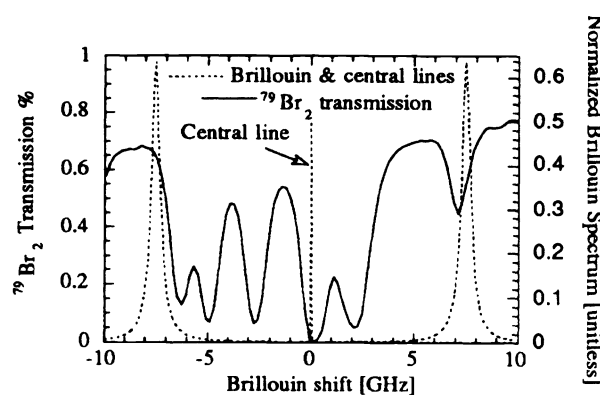


Fig. 5. Blocking filter : The ideal blocking filter should have a good transmission for the Brillouin lines and a total absorption for the central line of the backscattered spectrum. The absorption lines of  $^{79}\text{Br}_2$  provide a satisfactory blocking filter for an incident laser wavelength of 532.573806 nm.

## 4. PROSPECTS AND CONCLUSION

We have shown that an injection seeded pulsed laser can be used to obtain completely resolved measurements of Brillouin lines. This verifies that Brillouin shift data can be used to measure sound velocity as a function of depth in the ocean. A version of the edge technique for the measurement of the Brillouin shift has been outlined.

In comparison to other LIDAR receiver designs (using, for example, a Fabry-Perot) the  $\text{I}_2$  and/or  $\text{Br}_2$  absorption cells provide a simpler and more robust approach that requires no critical alignments. The optical efficiency of a receiver using the absorption lines discussed here, would be as good as that using a Fabry-Perot based receiver. A systematic search for other combinations of transmitter wavelength and absorption lines is underway. This includes direct experimental examination and comparison of the absorption lines of  $^{127}\text{I}_2$ ,  $^{129}\text{I}_2$ ,  $^{79}\text{Br}_2$ , and  $^{81}\text{Br}_2$ .

## 5. ACKNOWLEDGMENTS

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